Q1

```c
#include<stdio.h>
void main()
{
    int principal,rate,time;
    float simple_interest;
    printf("enter the value of principal:");
    scanf("%d", &principal);
    printf("enter the value of rate:");
    scanf("%d", &rate);
    printf("enter the value of time:");
    scanf("%d", &time);
    simple_interest=(principal*rate*time)/100.0;
    printf("the value of sinmple interest is:%f",simple_interest);
}
```

output

Problems Output Debug Console Terminal Ports

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 1file.c -o 1file } ; if ($?) {
 .\1file }
enter the value of principal:1000
enter the value of rate:2
enter the value of time:3
the value of sinmple interest is:60.000000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q2

```c
#include<stdio.h>
void main()
{
    float CS,MATHS,PHYSICS,CHEMISTRY,ENGLISH,total_marks;
    float percentage;
    printf("enter the marks of CS:");
    scanf("%f", &CS);
    printf("enter the marks of maths:");
    scanf("%f", &MATHS);
    printf("enter the marks of chemistry:");
    scanf("%f", &CHEMISTRY);
    printf("enter the marks of english:");
    scanf("%f", &ENGLISH);
    printf("enter the marks of physics:");
    scanf("%f", &PHYSICS);
    total_marks=CS+MATHS+PHYSICS+CHEMISTRY+ENGLISH;
    printf("total marks of student:%f\n",total_marks);
    percentage=(total_marks/500.0)*100.0;
    printf("percentage of student:%f",percentage);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 2file.c -o 2file } ; if ($?) {
 .\2file }
enter the marks of CS:92
enter the marks of maths:91
enter the marks of chemistry:99
enter the marks of english:95
enter the marks of physics:89
total marks of student:466.000000
percentage of student:93.199997
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q3

```c
#include<stdio.h>
void main()
{
    int gross_salary,basic_salary,house_rent_allowance,other_allowance,bonus;
    printf("basic salary of an employee is");
    scanf("%d", &basic_salary);
    printf("house rent allowance is");
    scanf("%d", &house_rent_allowance);
    printf("other allowance for employee is");
    scanf("%d", &other_allowance);
    printf("bonus earned by an employee is");
    scanf("%d", &bonus);
    gross_salary=basic_salary+house_rent_allowance+other_allowance+bonus;
    printf("gross salary of an employee= %d",gross_salary);

}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 3file.c -o 3file } ; if ($?) {
 .\3file }
basic salary of an employee is50000
house rent allowance is2000
other allowance for employee is5000
bonus earned by an employee is1000
gross salary of an employee= 58000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q4

```c
#include<stdio.h>
void main()
{
    float F,C; //F represents temperature in fahrenheit &
    // C represents temperature in celcius
    printf("temperature in fahrenheit:");
    scanf("%f", &F);
    C=(F-32.0)*(5.0/9.0);
    printf("temperature in celcius:%f",C);

}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 4file.c -o 4file } ; if ($?) {
 .\4file }
temperature in fahrenheit:212
temperature in celcius:100.000000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q5

```c
#include<stdio.h>
void main()
{
    int a,b,c;
    printf("enter value of a:");
    scanf("%d", &a);
    printf("enter value of b:");
    scanf("%d", &b);
    c=a;//swapping using third variable
    a=b;
    b=c;
    printf("after swapping:\n");
    printf("value of a=%d\n",a);
    printf("value of b=%d\n",b);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 5file.c -o 5file } ; if ($?) {
 .\5file }
enter value of a:5
enter value of b:10
after swapping:
value of a=10
value of b=5
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q6

```c
#include<stdio.h>
void main()
{
    int a,b;
    printf("enter the value of a=");
    scanf("%d", &a);
    printf("enter the  value of b=");
    scanf("%d", &b);
    a=a+b;
    b=a-b;
    a=a-b;
    printf("after swapping:\n");
    printf("value of a=%d\n",a);
    printf("value of b=%d",b);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 6file.c -o 6file } ; if ($?) {
 .\6file }
enter the value of a=2
enter the  value of b=1
after swapping:
value of a=1
value of b=2
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q7

```c
#include<stdio.h>
void main()
{
    float h,b,A;//where h is height & b is base of the triangle &
       // A represents area of the triangle
    printf("base of the triangle=");
    scanf("%f", &b);
    printf("height of the triangle=");
    scanf("%f", &h);
    A=(1.0/2.0)*b*h;
    printf("area of the triangle=%f",A);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 7file.c -o 7file } ; if ($?) {
 .\7file }
base of the triangle=4
height of the triangle=5
area of the triangle=10.000000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q8

```c
#include<stdio.h>
void main()
{
    int seconds,days,hours,minutes;
    printf("period of revolution of earth in seconds:");
    scanf("%d", &seconds);
    days=seconds/86400;   //number of seconds per day
    hours=seconds/3600;
    minutes=seconds/60;
    printf("period of revolution of earth in days:%d\n",days);
    printf("period of revolution of earth in hours:%d\n",hours);
    printf("period of revolution of earth in minutes:%d\n",minutes);

}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 8file.c -o 8file } ; if ($?) {
 .\8file }
period of revolution of earth in seconds:31558150
period of revolution of earth in days:365
period of revolution of earth in hours:8766
period of revolution of earth in minutes:525969
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q9

```c
#include<stdio.h>
void main()
{
    int hours,min,sec,total_seconds;
    printf("enter the hours:");
    scanf("%d", &hours);
    printf("enter the minutes:");
    scanf("%d", &min);
    printf("enter the seconds:");
    scanf("%d", &sec);
    total_seconds=(hours*3600)+(min*60)+(sec);
    printf("total time in seconds=%d",total_seconds);
}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 9file.c -o 9file } ; if ($?) {
 .\9file }
enter the hours:2
enter the minutes:15
enter the seconds:30
total time in seconds=8130
```

Q10

```c
#include<stdio.h>
void main()
{
    float M,P,C,E;
    float CM; /*where CM is for cut off mark
                    M for marks in maths out of 200
                    P for marks in physics out of 200
                    C for marks in chemistry out of 200
                    E for marks in entrance exam out of 100*/
    printf("marks in mathematics:");
    scanf("%f", &M);
    printf("marks in physics:");
    scanf("%f", &P);
    printf("marks in chemistry:");
    scanf("%f", &C);
    printf("marks in entrance examination:");
    scanf("%f", &E);
    CM=(M/2)+(P/2)+(C/2)+(E);
    printf("cut off mark of a student=%f",CM);

}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 10file.c -o 10file } ; if ($?)
 { .\10file }
marks in mathematics:180
marks in physics:178
marks in chemistry:190
marks in entrance examination:80
cut off mark of a student=354.000000
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```

Q11

```c
#include<stdio.h>
void main()
{
    int total_sec,hours,minutes,seconds;
    printf("total time in seconds:");
    scanf("%d", &total_sec);
    hours=total_sec/3600;
    total_sec=total_sec%3600;
    minutes=total_sec/60;
    seconds=total_sec%60;
    printf("time format(hours.min.sec):%d.%d.%d",hours,minutes,seconds);



}
```

output

```
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang> cd "c:\Users\pksir\OneDrive\Desktop\IOP_C_lang\" ; if ($?) { gcc 11file.c -o 11file } ; if ($?)
 { .\11file }
total time in seconds:8130
time format(hours.min.sec):2.15.30
PS C:\Users\pksir\OneDrive\Desktop\IOP_C_lang>
```